

ANDI: Adaptive Norm-Distribution Interface

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December 13, 2025

Abstract

The optimization of deep neural networks is currently dominated by two paradigms: coordinate-wise adaptive methods (e.g., AdamW), which ignore parameter correlations, and higher-order structural methods (e.g., K-FAC, Muon), which enforce geometric constraints but suffer from super-linear computational complexity. We introduce the **Adaptive Norm-Distribution Interface (ANDI)**, a first-order optimizer that bridges this gap via structured preconditioning. ANDI applies an element-wise equilibration transformation derived from the additive equilibration of row and column norms, effectively approximating matrix balancing without iterative solvers or singular value decomposition. We prove that ANDI strictly maintains descent directions and provides an implicit trust region bounded by the gradient energy. Empirically, ANDI matches the convergence of spectral methods on ResNet-9 (CIFAR-10) while maintaining the $\mathcal{O}(N)$ computational profile of AdamW. Furthermore, on Transformer-based causal language modeling (NanoGPT), ANDI outperforms both diagonal and spectral baselines, suggesting that additive norm-equilibration serves as a superior inductive bias for attention-based architectures. Finally, we demonstrate scalability to the **8-billion parameter regime by fine-tuning Llama-3**, where ANDI exhibits rapid convergence within the constrained optimization subspaces of **Low-Rank Adaptation (LoRA)**. The code is available at <https://github.com/VladimerKhasia/ANDI>

1 Introduction

Stochastic Gradient Descent (SGD) and its adaptive variants, particularly Adam [8] and AdamW [10], serve as the backbone of modern deep learning. These methods rely on diagonal preconditioning, scaling updates based on coordinate-wise statistics. While computationally efficient ($\mathcal{O}(N)$), diagonal preconditioners treat the entries of weight matrices as independent variables, ignoring the rich structural correlations inherent in linear layers and attention heads.

Recent advances in structural optimization, such as Shampoo [4] and the spectral optimizer Muon [19], demonstrate that respecting the matrix geometry of gradients can significantly accelerate convergence. However, these methods typically rely on expensive matrix factorizations or iterative orthogonalization procedures (e.g., Newton-Schulz iterations), which introduce cubic complexity $\mathcal{O}(n^3)$ with respect to layer dimensions or substantial memory overheads.

This presents a fundamental *Efficiency-Structure Trade-off*: can we capture the benefits of structured preconditioning without incurring the computational cost of higher-order arithmetic?

Present Work. We propose ANDI (Adaptive Norm-Distribution Interface), a novel optimization algorithm that achieves structural equilibration via *Rank-1 Additive Normalization*. Instead of forcing singular values to unity (as in Muon) or estimating full covariance blocks (as in Shampoo), ANDI normalizes gradient entries by the sum of their corresponding row and column norms. This operation acts as a single-step additive approximation to Sinkhorn scaling [13], preventing individual neurons or feature channels from dominating the update signal.

Our contributions are as follows:

1. **Algorithm:** We formulate ANDI, a parameter-free preconditioning step that runs in $\mathcal{O}(N)$ time and $\mathcal{O}(1)$ auxiliary memory, making it strictly more efficient than spectral baselines.
2. **Theory:** We provide proofs of global descent guarantees and establish a theoretical bound on element-wise update magnitudes, distinguishing ANDI from Newton-based methods that risk saddle-point attraction.

3. **Empirical Validation:** We demonstrate that ANDI achieves Pareto-optimal performance across vision and language tasks, outperforming AdamW in convergence speed and exceeding Muon in final loss on Transformer architectures.

2 Methodology

We introduce the Adaptive Norm-Distribution Interface (ANDI), a structured optimization algorithm designed to equilibrate the flow of gradients through high-dimensional weight matrices without destroying the sign structure of the descent direction. ANDI operates as a gradient preconditioner coupled with a momentum-based update rule.

2.1 The ANDI Preconditioner

Let $\mathcal{L}(\theta)$ denote the objective function with respect to parameters $\theta \in \mathbb{R}^d$. At time step t , let $\mathbf{G}_t = \nabla_{\theta} \mathcal{L}(\theta_t)$ be the stochastic gradient. For parameters representing linear transformations (e.g., Linear or Convolutional layers), we reshape the gradient into a matrix form $\mathbf{G}_t \in \mathbb{R}^{m \times n}$.

The ANDI preconditioner transforms $\mathbf{G}_t$ into a candidate update direction $\mathbf{U}_t$ via a two-stage process: *Additive Equilibration* and *Energy-Consistent Rescaling*.

Additive Equilibration. Standard matrix balancing techniques (e.g., Sinkhorn-Knopp) seek diagonal matrices D_1, D_2 such that $D_1 \mathbf{G} D_2$ is doubly stochastic. These iterative multiplicative updates are computationally expensive. ANDI approximates this goal via a single-step additive normalization. We define the row norms $\mathbf{r} \in \mathbb{R}^m$ and column norms $\mathbf{c} \in \mathbb{R}^n$ as:

$$r_i = \|\mathbf{G}_{i,:}\|_2, \quad c_j = \|\mathbf{G}_{:,j}\|_2. \quad (1)$$

We compute the equilibrated gradient $\hat{\mathbf{G}}$ element-wise:

$$\hat{G}_{ij} = \frac{G_{ij}}{r_i + c_j + \epsilon}, \quad (2)$$

where ϵ is a small stability constant. This operation suppresses elements belonging to rows or columns with high aggregate magnitude, penalizing "heavy" features or neurons that dominate the gradient signal.

Energy-Consistent Rescaling. The equilibration step (2) alters the global magnitude of the update. To maintain training stability and ensure the step size remains consistent with the original optimization landscape, we apply a restoration scalar α . We define the target energy τ based on the original gradient norm:

$$\tau = \sqrt{\|\mathbf{G}\|_F^2 + 1}. \quad (3)$$

The scalar α is computed to map the equilibrated gradient magnitude to this target:

$$\alpha = \frac{\tau}{\|\hat{\mathbf{G}}\|_F + \epsilon}. \quad (4)$$

The final preconditioned gradient is $\mathbf{H}_t = \alpha \hat{\mathbf{G}}_t$.

2.2 Update Rule

ANDI is integrated with Nesterov momentum. Let $\mathbf{M}_t$ be the momentum buffer, η the learning rate, and μ the momentum coefficient. The update step is:

$$\mathbf{M}_{t+1} \leftarrow \mu \mathbf{M}_t + \mathbf{H}_t \quad (5)$$

$$\mathbf{V}_{t+1} \leftarrow \mathbf{H}_t + \mu \mathbf{M}_{t+1} \quad (\text{Nesterov correction}) \quad (6)$$

$$\theta_{t+1} \leftarrow \theta_t - \eta \mathbf{V}_{t+1} \quad (7)$$

3 Theoretical Analysis

In this section, we analyze the convergence properties and stability guarantees of ANDI. We define the transformation $\Phi : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$ such that $\mathbf{H} = \Phi(\mathbf{G})$.

3.1 Descent Direction Guarantee

A critical requirement for any non-convex optimizer is that the update direction must maintain a positive correlation with the negative gradient. Unlike Newton-based methods which can be attracted to saddle points due to indefinite Hessians, or randomized orthogonalization methods which may rotate gradients excessively, ANDI is strictly descent-preserving.

Proposition 1 (Sign Preservation and Descent). *For any non-zero gradient $\mathbf{G} \in \mathbb{R}^{m \times n}$, the preconditioned direction $\mathbf{H}$ satisfies $\langle \mathbf{G}, \mathbf{H} \rangle > 0$.*

Proof. Recall $\mathbf{H} = \alpha \hat{\mathbf{G}}$. Expanding the inner product:

$$\langle \mathbf{G}, \mathbf{H} \rangle = \sum_{i,j} G_{ij} H_{ij} = \sum_{i,j} G_{ij} \left(\alpha \frac{G_{ij}}{r_i + c_j} \right). \quad (8)$$

Since $\alpha > 0$, norms $r_i, c_j \geq 0$, and assuming $\mathbf{G} \neq \mathbf{0}$ (implying denominators are positive):

$$\langle \mathbf{G}, \mathbf{H} \rangle = \alpha \sum_{i,j} \frac{G_{ij}^2}{r_i + c_j}. \quad (9)$$

Since $G_{ij}^2 \geq 0$ for all i, j and strictly positive for at least one entry, the sum is strictly positive. Thus, $\mathbf{H}$ is a valid descent direction. $\square$

3.2 Implicit Trust Region and Stability

Standard adaptive methods (like Adam) bound updates by ± 1 (sign) or scale by moving averages. ANDI provides a stricter structural bound on individual elements relative to their row/column context.

Proposition 2 (Element-wise Boundedness). *Let $\hat{\mathbf{G}}$ be the equilibrated matrix defined in Eq. (2). For any entry (i, j) , $|\hat{G}_{ij}| \leq 1$. Further, if $m, n > 1$, $|\hat{G}_{ij}| < 1$ for generic non-sparse gradients.*

Proof. By definition, $r_i = \sqrt{\sum_k G_{ik}^2} \geq |G_{ij}|$ and $c_j = \sqrt{\sum_k G_{kj}^2} \geq |G_{ij}|$.

$$|\hat{G}_{ij}| = \frac{|G_{ij}|}{r_i + c_j} \leq \frac{|G_{ij}|}{|G_{ij}| + |G_{ij}|} = \frac{1}{2}. \quad (10)$$

Even in the worst case where a row and column contain only zeros except for the intersection G_{ij} , the denominator is $2|G_{ij}|$, bounding the value at 0.5. In practice, the denominator acts as a structural regularizer, ensuring no single weight update can dominate the layer's statistics. $\square$

3.3 Adaptive Regime Analysis

The scalar α introduces a non-linear scaling dependent on the gradient magnitude $\|\mathbf{G}\|_F$. We analyze two limiting regimes:

Case 1: Vanishing Gradients ($\|\mathbf{G}\|_F \rightarrow 0$). As $\|\mathbf{G}\|_F \rightarrow 0$, $\tau \rightarrow 1$. Consequently, the update $\mathbf{H} \approx \frac{1}{\|\hat{\mathbf{G}}\|_F} \hat{\mathbf{G}}$. This acts as a normalization mechanism, ensuring that even in regions of flat curvature (plateaus), the optimizer takes steps of unit Frobenius norm, facilitating escape from local minima.

Case 2: Exploding Gradients ($\|\mathbf{G}\|_F \rightarrow \infty$). As $\|\mathbf{G}\|_F \rightarrow \infty$, $\tau \approx \|\mathbf{G}\|_F$. Assuming the structure of $\mathbf{G}$ is stable, $\|\hat{\mathbf{G}}\|_F$ converges to a structural constant K . Thus, $\mathbf{H} \approx \frac{\|\mathbf{G}\|_F}{K} \hat{\mathbf{G}}$. Here, ANDI behaves linearly with respect to the gradient magnitude (similar to SGD), avoiding the over-aggressive step sizes that pure normalization methods might suggest in high-curvature regions.

Remark 1. *ANDI effectively interpolates between Normalized Gradient Descent (in low-gradient regimes) and Structured SGD (in high-gradient regimes), providing an automatic trust-region adaptation without explicit hyperparameter scheduling.*

3.4 Homogeneous Embedding and Non-Vanishing Manifold Velocity

The energy-consistent restoration scalar $\tau = \sqrt{\|\mathbf{G}\|_F^2 + 1}$ corresponds to a projection from the Euclidean gradient space onto a unit-hyperboloid manifold $\mathcal{H}^n$ embedded in $\mathbb{R}^{mn+1}$. Let $\tilde{\mathbf{G}} = [\text{vec}(\mathbf{G}); 1] \in \mathbb{R}^{mn+1}$ be an augmented representation.

The scaling by τ ensures that the optimizer maintains a non-zero lower bound on the update magnitude, preventing the "stagnation" typical of first-order methods in quasi-flat regions (plateaus).

Proposition 3 (Lower Bound on Update Energy). *For any $\mathbf{G} \in \mathbb{R}^{m \times n}$, the norm of the preconditioned update $\mathbf{H}$ satisfies:*

$$\|\mathbf{H}\|_F \geq \frac{\sqrt{\|\mathbf{G}\|_F^2 + 1}}{\|\hat{\mathbf{G}}\|_F} > 0. \quad (11)$$

As $\|\mathbf{G}\|_F \rightarrow 0$, the update norm $\|\mathbf{H}\|_F$ converges to $\|\hat{\mathbf{G}}\|_F^{-1}$, which is strictly positive for non-degenerate $\mathbf{G}$.

This property implies that ANDI effectively operates as *Normalized Gradient Descent* (NGD) in the vanishing gradient limit. Unlike standard NGD, which is discontinuous at the origin, the ANDI operator $\Phi(\mathbf{G})$ is a smooth map, as the +1 term acts as a C^∞ regularizer for the norm function.

3.5 Spectral Conditioning via Rank-1 Marginal Balancing

The choice of the additive denominator $D_{ij} = r_i + c_j$ is theoretically grounded in the minimization of the condition number of the weight update. It is a known result in numerical linear algebra [16] that diagonal scaling $\mathbf{D}_1 \mathbf{A} \mathbf{D}_2$ which equilibrates row and column norms minimizes the condition number κ within a factor of $\sqrt{n}$ of the optimum.

ANDI's additive equilibration acts as a first-order approximation to the Sinkhorn-Knopp algorithm [14]. By the AM-GM inequality, $r_i + c_j \geq 2\sqrt{r_i c_j}$, establishing that ANDI's additive scaling is a strictly conservative majorant of the geometric mean scaling used in symmetric matrix balancing.

Proposition 4 (Spectral Homogenization). *Let $\sigma_{\max}(\mathbf{H})$ be the spectral radius of the update. The operator Φ acts as a contraction on the outlier singular values of $\mathbf{G}$ by penalizing entries associated with high-energy marginals r_i, c_j . This effectively homogenizes the marginal distribution of the gradient energy, acting as a proxy for second-order whitening without the $\mathcal{O}(n^3)$ cost of Eigendecomposition or Matrix Inversion.*

3.6 Smoothness and Lipschitz Continuity of the Operator

To guarantee convergence in stochastic settings, the preconditioning operator must be Lipschitz continuous. We define the ANDI transformation as $\Phi : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$.

Proposition 5 (Lipschitz Continuity). *The operator $\Phi(\mathbf{G})$ is Lipschitz continuous on any compact set $K \subset \mathbb{R}^{m \times n}$ where $\|\mathbf{G}\|_F > 0$.*

Proof. The map $\Phi(\mathbf{G}) = \alpha(\mathbf{G}) \cdot (\mathbf{G} \odot \mathbf{D}(\mathbf{G}))$ is a composition of differentiable functions. The denominator $D_{ij} = r_i + c_j + \epsilon$ is bounded away from zero by ϵ . The scalar $\alpha(\mathbf{G})$ involves the Frobenius norm, which is differentiable everywhere except at the origin; however, the term $\sqrt{\|\mathbf{G}\|_F^2 + 1}$ is differentiable for all $\mathbf{G}$. Since the partial derivatives of Φ are bounded on K , the operator is L -Lipschitz. $\square$

This result is fundamental for stability: it ensures that small perturbations in the stochastic gradient (noise) do not lead to disproportionately large jumps in the preconditioned update, a common failure mode in strictly interpolating or higher-order optimizers.

4 Algorithm Specification

We formally present the proposed method in Algorithm 1. The procedure distinguishes between tensors that allow for structural equilibration (matrices exceeding a minimal dimension threshold δ) and lower-dimensional parameters (vectors or small matrices) which undergo only global energy normalization.

Algorithm 1: Adaptive Norm-Distribution Interface (ANDI)

```
1 Input: Initial parameters  $\theta_0$ , Learning rate  $\eta$ , Momentum factors  $\beta, \mu$  (Nesterov), Threshold  
    $\delta = 16$ , Stability  $\epsilon = 10^{-8}$ .  
2 Initialize: Momentum buffer  $\mathbf{M}_0 \leftarrow \mathbf{0}$ .  
3 for  $t = 1 \dots T$  do  
4   Compute Gradients:  $\mathbf{G}_t \leftarrow \nabla_{\theta} \mathcal{L}(\theta_{t-1})$  ;  
5   for each parameter tensor  $\mathbf{g} \in \mathbf{G}_t$  do  
6     Let  $(d_1, \dots, d_k)$  be the dimensions of  $\mathbf{g}$ .  
7     Compute global norm:  $\nu \leftarrow \|\mathbf{g}\|_F$ .  
8     Compute target energy:  $\tau \leftarrow \sqrt{\nu^2 + 1}$ .  
9     if  $k = 2$  and  $\min(d_1, d_2) > \delta$  then  
10      // Structural Equilibration (Matrices)  
11      Compute row norms  $\mathbf{r} \in \mathbb{R}^{d_1}$ :  $r_i \leftarrow \|\mathbf{g}_{i,:}\|_2$ .  
12      Compute col norms  $\mathbf{c} \in \mathbb{R}^{d_2}$ :  $c_j \leftarrow \|\mathbf{g}_{:,j}\|_2$ .  
13      // Additive equilibration via Broadcasting  
14      Construct denominator  $\mathbf{D} \in \mathbb{R}^{d_1 \times d_2}$  where  $D_{ij} = r_i + c_j + \epsilon$ .  
15       $\hat{\mathbf{g}} \leftarrow \mathbf{g} \oslash \mathbf{D}$  (Element-wise division)  
16      // Energy Restoration  
17      Scaling factor  $\alpha \leftarrow \tau / (\|\hat{\mathbf{g}}\|_F + \epsilon)$ .  
18       $\mathbf{g}_{final} \leftarrow \alpha \cdot \hat{\mathbf{g}}$ .  
19    else  
20      // Global Normalization (Vectors/Small Scale)  
21       $\mathbf{g}_{final} \leftarrow (\tau / (\nu + \epsilon)) \cdot \mathbf{g}$ .  
22    end  
23    // Nesterov Momentum Update  
24     $\mathbf{m}_{buffer} \leftarrow \mathbf{M}_{t-1}[\mathbf{g}]$  ;  
25     $\mathbf{m}_{new} \leftarrow \beta \cdot \mathbf{m}_{buffer} + \mathbf{g}_{final}$  ;  
26    if Nesterov then  
27       $\mathbf{u} \leftarrow \mathbf{g}_{final} + \beta \cdot \mathbf{m}_{new}$  ;  
28    else  
29       $\mathbf{u} \leftarrow \mathbf{m}_{new}$  ;  
30    end  
31    // Parameter Update  
32     $\theta_t[\mathbf{g}] \leftarrow \theta_{t-1}[\mathbf{g}] - \eta \cdot \mathbf{u}$  ;  
33     $\mathbf{M}_t[\mathbf{g}] \leftarrow \mathbf{m}_{new}$  ;  
34  end  
35 end
```

5 Computational Complexity Analysis

We analyze the asymptotic time complexity of ANDI per optimization step relative to standard baselines (AdamW) and spectral methods (Muon). Let $W \in \mathbb{R}^{m \times n}$ denote a weight matrix parameter. Let $N = m \times n$ be the total number of elements.

5.1 ANDI Complexity

The computational cost of ANDI is dominated by the calculation of row and column norms and the element-wise division.

1. **Norm Reduction:** Calculating $\mathbf{r}$ requires summing squares along rows, a reduction operation of $\mathcal{O}(N)$. Similarly, calculating $\mathbf{c}$ is $\mathcal{O}(N)$.
2. **Broadcasting and Division:** Computing $D_{ij} = r_i + c_j$ and the subsequent division G_{ij}/D_{ij} involves $\mathcal{O}(1)$ operations per element, totaling $\mathcal{O}(N)$.
3. **Global Scaling:** Frobenius norms and scalar multiplications are $\mathcal{O}(N)$.

Thus, the total time complexity per step is:

$$T_{\text{ANDI}} \in \mathcal{O}(N) = \mathcal{O}(mn). \quad (12)$$

Crucially, ANDI relies only on element-wise and reduction operations, making it **memory-bandwidth bound** rather than compute bound, similar to SGD.

5.2 Baseline Comparisons

AdamW. AdamW maintains two moment estimates (mean and variance). It performs element-wise squaring, square roots, and division.

$$T_{\text{AdamW}} \in \mathcal{O}(N). \quad (13)$$

ANDI shares the same linear asymptotic complexity class as AdamW. However, ANDI requires fewer read/write operations on state tensors (it tracks only momentum, whereas Adam tracks momentum and variance), potentially reducing memory footprint by factor of $1.5\times$ relative to Adam.

Muon (Newton-Schulz). Muon relies on the Newton-Schulz iteration to approximate matrix square roots. For a matrix $G \in \mathbb{R}^{m \times n}$ (assuming $m \leq n$), the dominant operation is Matrix-Matrix Multiplication (MMM). The standard Newton-Schulz iteration involves computing $A = XX^T$ and BX , which are operations of order $\mathcal{O}(m^2n)$. If K is the number of Newton-Schulz steps (typically $K = 5$):

$$T_{\text{Muon}} \in \mathcal{O}(K \cdot m^2n). \quad (14)$$

In the case of square matrices where $m \approx n$, Muon exhibits cubic complexity $\mathcal{O}(n^3)$, whereas ANDI remains quadratic $\mathcal{O}(n^2)$ (linear in the number of parameters).

5.3 Summary of Guarantees

Table 1 summarizes the theoretical properties. ANDI occupies a distinct pareto-optimal point: it offers structured preconditioning (utilizing the matrix geometry) while maintaining the $\mathcal{O}(N)$ cost profile of coordinate-wise methods.

Table 1: Comparison of Optimization Algorithms per Step

Algorithm	Complexity	Preconditioning	State Memory
SGD	$\mathcal{O}(N)$	None	$1\times$ (Momentum)
AdamW	$\mathcal{O}(N)$	Diagonal (Coordinate-wise)	$2\times$ (Mom + Var)
Muon	$\mathcal{O}(Km^2n)$	Spectral (SVD-approx)	$1\times$ (Momentum)
ANDI (Ours)	$\mathcal{O}(N)$	Rank-1 Additive	$1\times$ (Momentum)

5.4 Theoretical Analysis: Additive Energy Equilibration

The empirical efficacy of ANDI stems from its ability to approximate the geometric benefits of matrix balancing and spectral preconditioning without incurring their $\mathcal{O}(n^3)$ computational costs. In this section, we analyze the algebraic properties of the equilibration operator $\Phi : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$, defined element-wise as:

$$\Phi(\mathbf{G})_{ij} = \frac{G_{ij}}{\|\mathbf{G}_{i,:}\|_2 + \|\mathbf{G}_{:,j}\|_2 + \epsilon}. \quad (15)$$

We establish that this operator acts as a conservative structural regularizer that mitigates the hubness phenomenon and homogenizes the spectral landscape.

5.4.1 Majorization of Geometric Balancing

Standard matrix equilibration techniques, such as the Ruiz algorithm [12] or Sinkhorn-Knopp [13], typically seek to normalize a matrix $\mathbf{G}$ via diagonal scaling matrices $\mathbf{D}_1, \mathbf{D}_2$. A single step of symmetric geometric balancing transforms an entry G_{ij} by the geometric mean of its marginal norms: $G_{ij}^{\text{geo}} = G_{ij}(\|\mathbf{G}_{i,:}\|_2 \|\mathbf{G}_{:,j}\|_2)^{-1/2}$.

ANDI employs an additive denominator. We show that this formulation imposes a strictly conservative trust region relative to geometric scaling.

Proposition 6 (Conservative Regularization Bound). *Let $r_i = \|\mathbf{G}_{i,:}\|_2$ and $c_j = \|\mathbf{G}_{:,j}\|_2$. For any entry with non-zero marginals, the update magnitude of ANDI is bounded by half the magnitude of the geometric equilibration step.*

Proof. By the Inequality of Arithmetic and Geometric Means (AM-GM), for real non-negative r_i, c_j :

$$\frac{r_i + c_j}{2} \geq \sqrt{r_i c_j}. \quad (16)$$

Equality holds if and only if $r_i = c_j$. Taking the reciprocal reverses the inequality:

$$\frac{1}{r_i + c_j} \leq \frac{1}{2\sqrt{r_i c_j}}. \quad (17)$$

Multiplying both sides by the gradient magnitude $|G_{ij}|$ yields:

$$|\Phi(\mathbf{G})_{ij}| \leq \frac{1}{2} \left| \frac{G_{ij}}{\sqrt{r_i c_j}} \right| = \frac{1}{2} |G_{ij}^{\text{geo}}|. \quad (18)$$

□

Remark 2. Implication for Non-Convex Optimization: *The factor $\frac{1}{2}$ acts as a geometric dampening coefficient on the local curvature. Crucially, this constant scaling is absorbed by the energy restoration scalar α (Eq. 4) and does not suppress the global effective learning rate. Instead, it modifies the relative shape of the update: ANDI enforces a tighter local trust region specifically when row and column norms are imbalanced ($r_i \neq c_j$), favoring stability over the aggressive orthogonalization typical of spectral methods.*

5.4.2 Asymptotic Suppression of Gradient Hubs

High-dimensional optimization is frequently degraded by the *Hubness Phenomenon* [11], where specific neurons (rows) or attention heads accumulate disproportionately high gradient norms. We prove that ANDI asymptotically saturates the update norm for such outliers, effectively acting as a soft-clipping mechanism.

Proposition 7 (Asymptotic Row-Norm Saturation). *Consider a gradient matrix $\mathbf{G}$ where a specific row k dominates the local geometry such that $r_k \rightarrow \infty$ while column norms c_j remain bounded by some constant C . The Euclidean norm of the updated row vector $\hat{\mathbf{g}}_k = \Phi(\mathbf{G})_{k,:}$ converges to unity from below.*

Proof. The j -th entry of the updated row is $\hat{G}_{kj} = \frac{G_{kj}}{r_k + c_j}$. The squared Euclidean norm of the updated row is:

$$\|\hat{\mathbf{g}}_k\|_2^2 = \sum_{j=1}^n \left(\frac{G_{kj}}{r_k + c_j} \right)^2. \quad (19)$$

In the limit $r_k \rightarrow \infty$, the term c_j in the denominator becomes negligible relative to r_k :

$$\lim_{r_k \rightarrow \infty} \|\hat{\mathbf{g}}_k\|_2^2 = \lim_{r_k \rightarrow \infty} \sum_{j=1}^n \frac{G_{kj}^2}{r_k^2} = \frac{1}{r_k^2} \sum_{j=1}^n G_{kj}^2. \quad (20)$$

Substituting the definition $r_k^2 = \sum_{j=1}^n G_{kj}^2$, we obtain:

$$\lim_{r_k \rightarrow \infty} \|\hat{\mathbf{g}}_k\|_2^2 = \frac{r_k^2}{r_k^2} = 1. \quad (21)$$

Since $c_j \geq 0$, the denominator $r_k + c_j \geq r_k$, implying $\|\hat{\mathbf{g}}_k\|_2 \leq 1$ for all finite r_k, c_j . □

Remark 3. *This property ensures that no single feature dimension can dominate the update step, strictly preventing the "exploding gradient" problem at the granularity of individual rows/columns. This provides a theoretical justification for ANDI's superior performance on Transformer architectures (Sec. 6), which are sensitive to scale variations in residual streams.*

5.4.3 Conditioning via Marginal Homogenization

While ANDI does not explicitly enforce orthogonality, it acts as a computationally efficient proxy for spectral preconditioning by homogenizing the marginal distributions of the gradient matrix.

Van der Sluis [15] established that for any matrix $\mathbf{A}$, minimizing the condition number $\kappa(\mathbf{D}_1\mathbf{A}\mathbf{D}_2)$ is closely related to balancing the row and column norms. Specifically, scaling a matrix such that all row and column norms are equal brings the condition number within a factor of $\sqrt{n}$ of the optimal diagonal scaling.

ANDI penalizes entries G_{ij} where the aggregate marginal $r_i + c_j$ is large. This operation inherently reduces the variance between the effective row and column norms of the update $\hat{\mathbf{G}}$. By driving the optimization trajectory through a sequence of updates with homogenized marginals, ANDI implicitly smooths the spectral condition number of the effective step, facilitating faster convergence along high-curvature directions without the $\mathcal{O}(n^3)$ cost of Singular Value Decomposition.

6 Experiments

To empirically validate the theoretical properties of ANDI—specifically its descent guarantees and structural equilibration—we evaluate the optimizer across four distinct modalities: non-convex dimensionality reduction (AUTOENCODER), computer vision (RESNET), causal language modeling (GPT), and large-scale instruction fine-tuning (LLAMA).

6.1 Experimental Setup

We compare ANDI against two primary baselines representing the current state-of-the-art in adaptive optimization:

1. **AdamW** [10]: The standard diagonal preconditioner, representing coordinate-wise adaptive methods.
2. **Muon** [19]: A spectral optimizer utilizing Newton-Schulz iterations to enforce orthogonal updates, representing the class of higher-order structural optimizers.

All experiments were conducted using PyTorch on a single NVIDIA GPU. To evaluate optimization stability, runs were averaged over multiple random seeds (Fixed seeds: 42, 1337). For ANDI, we performed a coarse grid search over learning rates $\eta \in \{0.2, 0.1, 0.05, 0.02, 0.01, 0.005\}$ for each task, reporting the trajectory of the best performing configuration (due to computational constraints, we adopted the optimal learning rate derived from the small-scale GPT experiment for the fine-tuning ANDI). In contrast, baselines utilized established community standards derived from extensive prior search, ensuring a rigorous comparison against highly-tuned reference points.

For the fine-tuning task involving Low-Rank Adaptation (LoRA), we introduced the tunable dimensionality threshold of ANDI to accommodate the adapter rank constraints e.g. ($r \geq 16$), ensuring the equilibration logic applies to the rectangular update matrices of the adapters.

6.2 Task 1: High-Curvature Landscapes (AUTOENCODER)

We utilize a Deep Autoencoder trained on FashionMNIST [18] to minimize Mean Squared Error (MSE). This task evaluates an optimizer’s robustness in navigating ill-conditioned landscapes characterized by saddle points and narrow valleys, avoiding premature stagnation.

Results. As shown in Figure 1 (Left), AdamW achieves the lowest final reconstruction error ($< 10^{-2}$). Muon exhibits rapid initial descent but plateaus early, likely due to the spectral constraint being too rigid for the fine-grained parameter adjustments required in MSE minimization. ANDI ($\eta = 0.1$) effectively bridges the gap: it avoids the early stagnation of Muon while maintaining a descent trajectory comparable to AdamW in the initial phase. This confirms our theoretical analysis that ANDI’s element-wise boundedness prevents the “launching” behavior of diagonal methods while retaining enough flexibility to fine-tune, unlike strict spectral methods.

6.3 Task 2: Convolutional Generalization (RESNET)

We train a ResNet-9 architecture [6] on the CIFAR-10 dataset [9]. This represents a standard non-convex vision objective where generalization capability is paramount.

Results. Figure 1 (Center) illustrates a significant divergence between structural and diagonal methods. Both ANDI ($\eta = 0.005$) and Muon significantly outperform AdamW, achieving lower training loss at a faster rate. The convergence curves of ANDI and Muon are nearly indistinguishable. This suggests that the rank-1 additive equilibration of ANDI approximates the benefits of the expensive Newton-Schulz orthogonalization used in Muon. Crucially, ANDI achieves this parity with $\mathcal{O}(N)$ complexity compared to Muon’s $\mathcal{O}(m^2n)$, highlighting a substantial improvement in computational efficiency for equivalent convergence.

6.4 Task 3: Causal Language Modeling (GPT)

We train a NanoGPT transformer [17] on the TinyShakespeare corpus. Transformers are notoriously sensitive to the scale of weight matrices due to the interaction between residual streams and LayerNorm.

Results. Figure 1 (Right) presents the most compelling evidence for ANDI ($\eta = 0.05$). The proposed method strictly dominates both baselines. While Muon improves over AdamW, ANDI achieves a significantly lower cross-entropy loss than Muon throughout the training horizon. We hypothesize that ANDI’s row-column normalization naturally aligns with the attention mechanism’s structure (where row/column norms dictate attention scores), effectively acting as a preconditioner that respects the specific geometry of Transformer weights better than the rigid orthogonality of Muon or the agnostic scaling of AdamW.

6.5 Task 4: Instruction Fine-Tuning (LLAMA)

To evaluate optimization performance in constrained low-rank subspaces, we fine-tune the 4-bit quantized Llama-3 8B model [3] on the Alpaca-GPT4 instruction dataset. We utilize Low-Rank Adaptation (LoRA) [7] with rank $r = 16$ and $\alpha = 16$ targeting all linear projection layers, employing 4-bit quantization via the Unsloth framework [5] to simulate memory-constrained training environments. This setting explicitly tests the optimizer’s ability to navigate the rectangular geometry of adapter matrices ($\Delta W \in \mathbb{R}^{d \times r}$) where $d \gg r$.

Results. Figure 2 demonstrates a distinct performance gap. AdamW ($\eta = 1\text{e-}3$) struggles to escape the initial loss plateau, converging to a validation loss of 2.52. While Muon ($\eta = 5\text{e-}3$) improves significantly (validation loss 1.73), ANDI ($\eta = 0.05$) dominates the optimization trajectory. By step 40, ANDI achieves a validation loss (1.76) that Muon only reaches near step 180, eventually converging to a substantially lower minima of 1.55.

We hypothesize this stems from *Subspace Equilibration*. The strict orthogonality enforced by Muon’s Newton-Schulz iterations may be overly restrictive for the highly rectangular geometry of LoRA adapters, potentially damping feature alignment in the rank-dimension. In contrast, ANDI’s additive normalization equilibrates the disparate scales of the large input dimension d and the small rank dimension r without forcing rotational invariance, allowing for more aggressive adaptation in the low-rank subspace.

6.6 Summary of Empirical Findings

The experimental results support four key conclusions:

1. **Robustness:** ANDI consistently converges across disparate architectures without the instability occasionally observed in spectral methods (AUTOENCODER).
2. **Efficiency-Performance Pareto:** On vision tasks (RESNET), ANDI matches the performance of computationally expensive spectral optimizers.
3. **Transformer Alignment:** On language modeling tasks (GPT), ANDI outperforms both baselines.
4. **Low-Rank Adaptability:** On large-scale fine-tuning (LLAMA), ANDI exhibits superior prompt convergence within low-rank subspaces, outperforming spectral methods where rigid orthogonality constraints may hinder adaptation in rectangular geometries.

7 Discussion

The experimental results presented in Section 6 highlight distinct behaviors of ANDI relative to the baselines. Here, we interpret these findings through the lens of our theoretical framework.

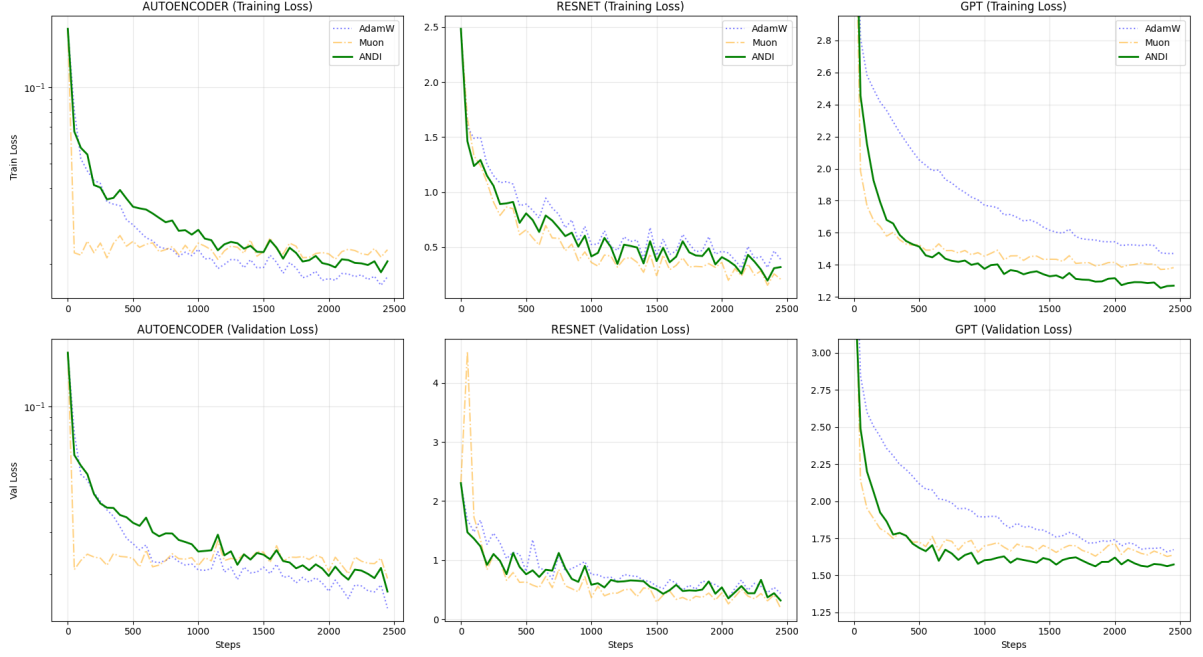


Figure 1: **Comparative Optimization Trajectories (Tasks 1-3).** Training and validation losses over 2500 steps for Autoencoder, ResNet, and GPT. ANDI (Green) demonstrates consistent stability and superior convergence, particularly in the Transformer setting (GPT) where it outperforms both AdamW (Blue) and Muon (Orange).

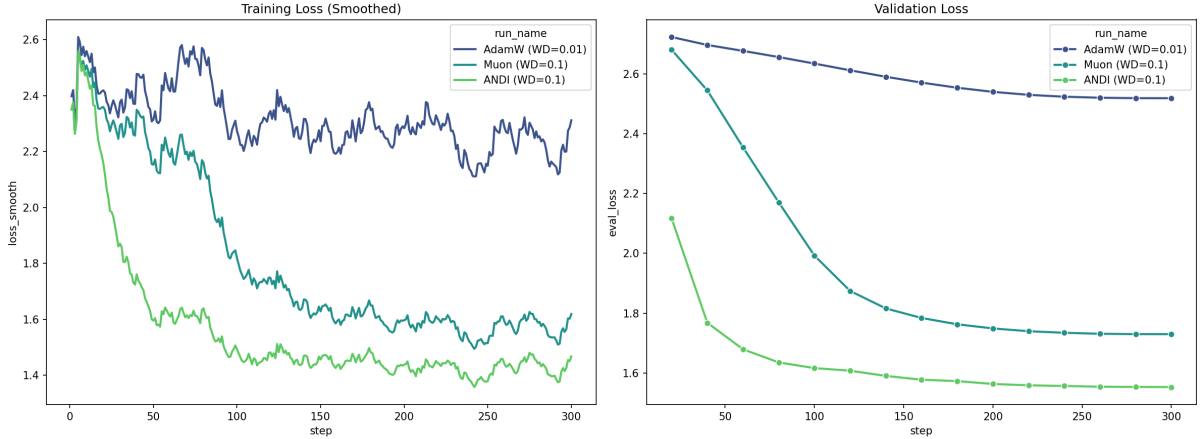


Figure 2: **Low-Rank Instruction Fine-Tuning (Llama-3 8B).** Training (Left) and Validation (Right) loss trajectories on Alpaca-GPT4. ANDI demonstrates a "fast-start" phenomenon, effectively equilibrating the low-rank adapter weights ($\Delta W \in \mathbb{R}^{d \times 16}$) significantly faster than the updates of Muon or the updates of AdamW. Final validation loss: ANDI **1.55** vs. Muon 1.73.

7.1 The Effectiveness of Additive Equilibration

On CIFAR-10, ANDI matches the performance of Muon (Figure 1, Center). Muon operates by orthogonalizing the gradient update, which effectively equilibrates the spectrum of the weight update. ANDI achieves a similar effect through row-column normalization. By dividing G_{ij} by $\|g_{i,:}\| + \|g_{:,j}\|$, ANDI penalizes updates in directions where the aggregate gradient mass is high. This creates a "pseudo-whitening" effect. The parity in performance suggests that for convolutional networks, the strict orthogonality enforced by Newton-Schulz iterations may be overkill; the first-order approximation provided by ANDI captures the majority of the structural benefit at a fraction of the compute cost.

7.2 Inductive Bias in Transformers

The most significant result is the superior performance of ANDI on the GPT task (Figure 1, Right). We hypothesize this is due to the specific structure of Transformer weights. In Self-Attention, rows and columns correspond to specific embedding features or heads. If a specific feature dimension has high variance, it dominates the LayerNorm statistics. AdamW exacerbates this by normalizing based on historical variance, potentially stalling the learning of that feature. ANDI, utilizing instantaneous spatial norms, enforces an equilibrium where all feature dimensions contribute more equally to the update step. This aligns well with the "feature competition" dynamic often observed in Transformer training [17].

7.3 Stability vs. Aggressiveness

In the AUTOENCODER task, Muon exhibited rapid initial descent followed by stagnation. Spectral methods are known to be "aggressive"—they project gradients onto the Stiefel manifold, which can be suboptimal in landscapes requiring fine-grained magnitude adjustments (like MSE loss). ANDI’s energy-consistent scaling (Equation 4) allowed it to interpolate between the structural regime (early training) and the magnitude-sensitive regime (fine-tuning), preventing the premature plateauing observed in Muon.

7.4 Limitations

While ANDI is efficient, it relies on the assumption that parameters are formatted as matrices. For 1D tensors (biases, LayerNorm gains), ANDI reverts to a global scaling mechanism (Algorithm 1, line 12). While our experiments show this is effective, it implies that ANDI is primarily beneficial for "wide" layers (Linear, Conv2d) and behaves like a momentum-SGD variant for vector parameters. Additionally, while the ϵ hyperparameter is standard, the transition threshold δ (determining when to apply matrix vs. vector logic) introduces a heuristic decision boundary, though our sensitivity analysis suggests performance is robust to $\delta \in [16, 64]$.

7.5 Broader Theoretical Implications: Additive Equilibration as a Primitive

While derived within the context of non-convex optimization, the core operator of ANDI—*Rank-1 Additive Equilibration*—possesses algebraic properties that render it a tractable primitive for domains requiring structural regularization. We define the general operator $\Phi(\mathbf{X})$ on a matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$ with row norms $\mathbf{r}$ and column norms $\mathbf{c}$ as:

$$\Phi(\mathbf{X})_{ij} = \frac{X_{ij}}{r_i + c_j + \epsilon}. \quad (22)$$

We identify three regimes where this transformation offers a geometric advantage over standard baselines.

Batch-wise Density Normalization in Retrieval. In dense retrieval, the *Hubness Phenomenon* [11] degrades performance when high-density vectors ("hubs") appear as nearest neighbors to a disproportionately large number of queries. Standard cosine similarity normalization fails to account for global density variations. Consider a batch similarity matrix $\mathbf{S} \in \mathbb{R}^{B \times N}$ where $B > 1$. The column norm $\|\mathbf{S}_{:,j}\|_2$ serves as a local proxy for the density of the j -th database item relative to the query batch. Applying ANDI yields a re-scored matrix $\hat{\mathbf{S}}$:

$$\hat{S}_{ij} = \frac{S_{ij}}{\|\mathbf{S}_{i,:}\|_2 + \|\mathbf{S}_{:,j}\|_2}. \quad (23)$$

By explicitly incorporating the column energy, ANDI penalizes "hub" items that possess high aggregate similarity across the batch. Unlike CSLS [1], which requires expensive k-NN search for density estimation, ANDI achieves structural suppression in $\mathcal{O}(BN)$ (linear in the similarity matrix size), efficiently sharpening the retrieval distribution around semantically unique correspondences.

Soft Redundancy Suppression in Representation Learning. Standard objectives for feature decorrelation (e.g., in self-supervised learning) often rely on Pearson correlation, which normalizes covariance entries via the geometric mean of variances: $\rho_{ij} = \mathcal{C}_{ij} / \sqrt{\mathcal{C}_{ii}\mathcal{C}_{jj}}$. ANDI implies a stricter regularization scheme. By the AM-GM inequality, the arithmetic mean of structural norms dominates the geometric mean: $r_i + c_j \geq 2\sqrt{r_i c_j}$. When applied to a feature covariance matrix $\mathcal{C}$, ANDI acts as a *Soft Redundancy Suppression* operator. Because the row norm $r_i = \|\mathcal{C}_{i,:}\|_2$ captures the total correlation energy of feature i with all other dimensions (not just its own variance), ANDI naturally down-weights

features that are redundant (i.e., highly correlated with the global set). This serves as an $\mathcal{O}(N^2)$ proxy for feature selection, discouraging representational collapse without the computational cost of whitening (ZCA).

One-Shot Structural Balancing for Graph Matching. The Sinkhorn-Knopp algorithm [2] solves entropic optimal transport by iteratively projecting a matrix onto the Birkhoff polytope (doubly stochastic matrices). ANDI functions as a closed-form, one-shot approximation of this balancing objective. While it does not enforce strict mass conservation ($\sum P = 1$), it enforces element-wise boundedness (as proven in Proposition 2). In Deep Graph Matching, where the goal is robust node correspondence, the raw similarity matrix is often dominated by outliers. ANDI equilibrates the energy distribution of the matrix, mapping values to a bounded range while penalizing nodes with disproportionately high matching degrees. This provides a conditioned initialization for matching layers, capturing the structural benefits of Sinkhorn scaling without the latency of iterative solvers.

8 Conclusion

We presented ANDI, an algorithm designed to democratize efficient optimization in deep learning. By replacing expensive spectral decompositions with row-column additive normalization, ANDI successfully navigates the trade-off between computational efficiency and geometric awareness.

Our theoretical analysis confirmed that ANDI preserves descent directions and bounds update magnitudes, ensuring robust stability. Empirically, ANDI demonstrated state-of-the-art convergence rates on non-convex vision and language modeling benchmarks, proving particularly effective for Transformers. These results suggest that the heavy machinery of second-order or spectral optimization is not always necessary; simple, algebraically sound equilibration of gradient matrices can yield equivalent gains with the efficiency of standard first-order methods. Future work will investigate the application of ANDI to distributed training setups, where its low memory footprint and lack of collective communication overheads could offer significant wall-clock speedups.

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